

Placing Nature

Interactive form-finding using computer vision and robotic arm collaboration for natural timber structure

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Abstract. The interplay between material, forces, and form is exemplified in an interactive design-build process. However, digital design and robotic fabrication often rely on deterministic workflows and precise geometry. This project investigates a dynamic design-build process that fosters creativity through human-robot collaboration. An interactive robotic fabrication method was developed, incorporating the YOLOv5 model and gesture tracking to create improvised natural structures. The robotic arm classified the wood elements selected by humans and identified potential placement points, integrating human input to refine the process until a natural timber structure emerged. Throughout the pick-and-place phases, humans and robots alternated roles in learning and construction. Ten participants constructed forty test mockups and ten final natural structures, followed by interviews and questionnaires about the process. Participants reflected on their experiences, gaining deeper insights into the detection model, the robot's performance in relation to existing structures, and the role of material constraints in shaping new forms. The findings underscore how humans adapt to physical feedback and machine operations to design with materiality and natural characteristics.

Keywords. Material-awareness, Form-finding, Robotic Fabrication, Human Computer Interaction, Computer Vision

1. Introduction

The acquisition of knowledge on materials, forces, and forms constitutes an interactive process of design and construction. Computer-aided design tools have transformed how we perceive and reshape our environments. Reflecting on approaches such as CNC, 3D printing, robotic assembly, and geometric modeling, it is essential to maintain adaptability, build on established relationships and rules, and transition from ambiguous to deliberate associations between objects (Mark, 2008). Digital design and robotic fabrication often rely on deterministic workflows and precise geometry, which

can constrain design space exploration and material understanding (Capunaman, 2020), as well as the exploration of forces and forms. However, design is inherently a non-linear process, particularly in its early phases, where accidents and material inaccuracies can inspire further development. Even without a fully formed mental image of the design, designers often improvise by manually building mockups with available materials. During physical modeling, gestures, and tool usage, creativity emerges through the improvised interaction between machines and tools (Pinochet, 2015). The connection between the human body, tools, and materials is crucial for establishing an interactive learning process that fosters design progress and generates new insights.

This paper explores a design space where designers can create new structural mockups using natural timber elements. Instead of designing tool paths by precisely positioning scanned timber branches, this study adopts a self-adjusting process influenced by gravity, allowing the structure to form iteratively and achieve self-stabilization. Both humans and machines follow predefined rules and engage in a progressive learning process to manipulate materials and gain practical experience. We introduce self-trained models developed using YOLOv5 (Ultralytics, 2020) and the hand gesture detection model (Lugaresi et al., 2019) to establish communication between humans and machines, facilitating the investigation of material and structural performance (Figure 1). The user testing results demonstrate the effectiveness of this approach for creating innovative natural timber structures through human-robot collaboration.

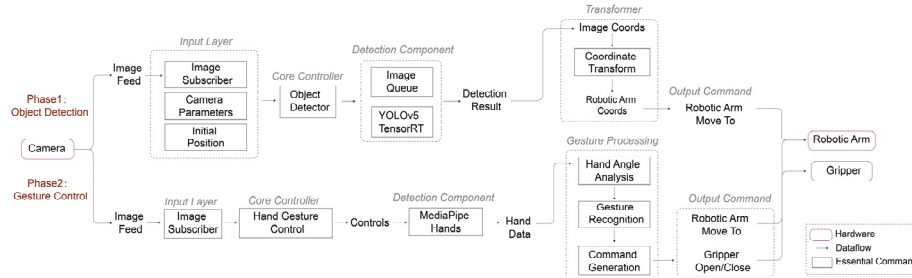


Figure 1. Pipeline

2. Related Work

In industry, robotic fabrication processes are typically based on predetermined tool paths generated for specific tasks, which cannot be adjusted once the machine begins operation. Recent studies on collaborative robotic construction have tested various materials to develop innovative forms and workflows. For instance, The Filament Winding study (Lauren Vasey et al., 2016) describes an interdisciplinary project that investigates collaboration between untrained humans and robots to construct a large-scale structure. Similarly, The Human Touch in Digital Fabrication (Lopez et al., 2016) introduces a hybridized method that combines human movements, lightweight composite materials, a handheld smart tool, and generative forms through an iterative feedback system. While these studies incorporate human participation in the design-build process, such involvement is primarily limited to the early stages of fabrication.

Notably, some researchers have sought to develop manufacture-aware and interactive fabrication systems that enable real-time feedback during machine operation. Certain studies have explored novel 3D printing workflows with instantaneous response capabilities. For example, The Responsive Spatial Print project (Im et al., 2018) introduced a closed-loop feedback system that employed material surveying and real-time scanning to detect local deviations from the digital model, adjusting the deposition process accordingly. Similarly, The Interactive Printing study (Capunaman, 2020) captured momentary changes in the fabricated artifact introduced by users to inform the digital representation. Other studies have incorporated body movements and gesture detection to control machine actions. For instance, The Making Gestures project (Pinochet, 2015) investigated methods for integrating the mind, body, and tools to transform design processes. It used body gestures to animate fabrication machines, establishing a dynamic dialogue that embraced ambiguity and unpredictability, thereby encouraging designers to engage deeply in creative and reflective processes. Lastly, the Wax form-making project (Lim, 2023) utilized AI-driven image recognition to capture designers' hand gestures and movements as an intuitive input method for controlling shape formation. The Enactive Robotics study (Garcia del Castillo Lopez, 2019) introduced a model that emphasized real-time responsiveness and immediate feedback, supporting experimentation, fluid thought, and creative exploration. In the context of digital fabrication, this model effectively enabled closed-loop control, incorporating real-time material feedback throughout the process. The study also proposed a user interface (UI)--based application that allowed both beginners and professionals to utilize robotic tools for design, demonstrating the effectiveness of robotic fabrication systems as universal design tools.

However, the dynamic learning process for both humans and robots remains largely underexplored. The progressive understanding and emergence of creativity in robotics, materiality, and form-finding processes within robotic fabrication are challenging to identify and evaluate. New interactive methods must be developed to investigate human-robot communication and its potential to generate novel forms and express materiality in fabrication.

3. Methodology

This paper explores an interactive robotic workflow (Figure 2) to investigate the form-finding process of natural timber structures using computer vision and object classification and detection. The robotic arm utilizes a self-trained object classification model developed with YOLOv5 (Ultralytics, 2020) to categorize timber branches based on their geometric characteristics. Timber branches were collected from yard trimmings, and a dataset was created using both these collected branches and online images. After training with the compiled dataset, the robotic arm can distinguish between three types of branches, each associated with unique structural behavior and constructability.

The interactive fabrication process (Figure 2) began after users built a foundation. The participant assisted the robotic arm in picking up a timber branch and guiding it along the y-axis while the robotic arm determined the x-axis and z-axis of the placement point. To achieve this, the robot was trained to learn from the interactive human build-test process using the YOLOv5 model (Ultralytics, 2020) to detect

potential placement points on the existing structure. This allowed the robot to identify the most suitable locations for branch additions by moving to the potential position on the elevation. After detection and elevation adjustment, the robotic arm fine-tuned its end effector's forward and backward movement along the y-axis based on hand gesture input from participants, implemented using the MediaPipe library (Lugaresi et al., 2019). The final drop location is determined collaboratively by the robot and human, reflecting a continuous calibration of force reactions through iterative processes. At the beginning of the user testing, participants were required to predict the location where the next branch would fall. Both the predicted position and the actual position of the timber branches were recorded. After the testing, participants were asked to provide feedback on the interactive fabrication process to reflect on the dynamics of human-robot collaboration.

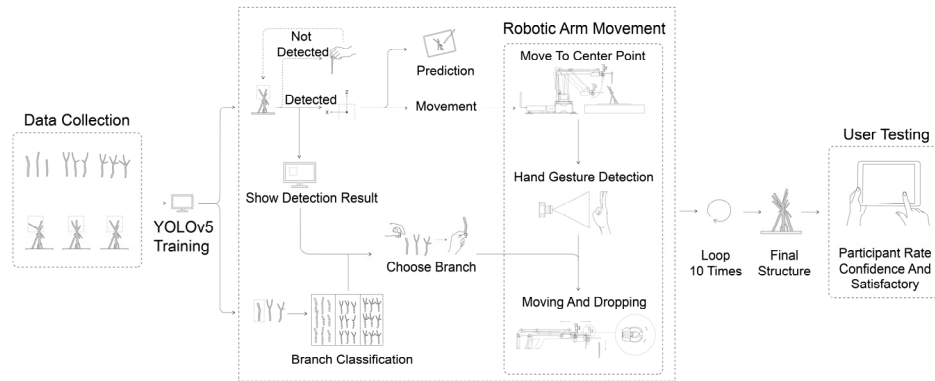


Figure 2 Robotic Interactive fabrication workflow

4. Interactive Form-Finding System

4.1 REAL-TIME DESIGN SPACE SET-UP

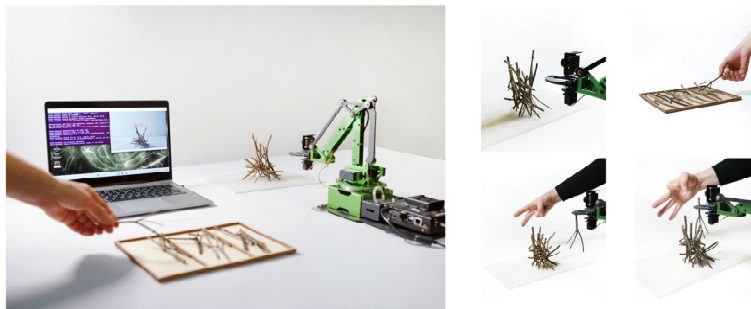


Figure 3 Design space setup

The study utilizes a desk robotic arm with 4+1 degrees of freedom, equipped with a specially designed 3D-printed gripper featuring a foam-covered curved groove to

securely hold naturally shaped timber branches. Throughout the testing, the robotic arm is positioned in front of a construction plate, where it adds branches to a human-built foundation with input from participants (Figure 3). Its camera is aligned with the vertical plane, while the gripper remains parallel to the horizontal plane. The structure is placed on a 3d -printed construction plate with grooves and bumps on its surface to ensure basic support for the timber structure. A remotely connected computer displaying the robotic arm's detection results is positioned on the left side. A preparation area containing different types of timber branches is situated nearby, allowing users to select the next element to add to the structure (Figure 3).

Participants stand beside the robotic arm and primarily observe the structure from a top-down view. They interact with the robotic arm via the camera, using hand gestures facilitated by the MediaPipe library (Lugaresi et al., 2019).

4.2 RAW TIMBER INVENTORY BUILDING AND CLASSIFICATION

For the preparation of the timber branch dataset, fallen timber branches were collected from natural environments, with diameters ranging from approximately 2 to 5 millimeters and lengths of around 150 millimeters. The branches were categorized into three types: stick shape, Y shape, and claw shape. This categorization is based on the branching patterns, which influence constructability and stability during the building process. Figure 4 illustrates typical examples of each branch type. The dataset contains a total of 600 photographs, including 500 self-taken photos and 100 images sourced from the internet.

YOLOv5 (Ultralytics, 2020) is used to train the data sets. After training with the timber branches dataset, the model can distinguish among three types of branches and achieves a confidence score of around 0.6-0.9 for both self-prepared branches and branches pictures from the internet.

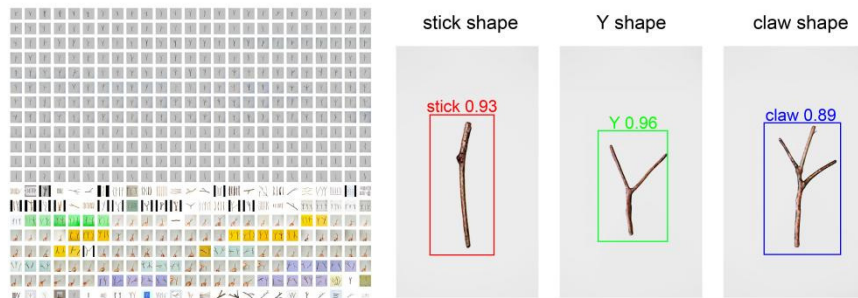


Figure 4 Timber branch dataset

4.3 MACHINE VISION AND POTENTIAL PLACEMENT CONFIGURATION DETECTION

Previous studies, such as Designing Natural Wood Log Structures (Wu & Kilian, 2018), utilized 3D scanning to analyze log surfaces and trained convolutional neural networks (CNNs) to identify picking points and positions. In contrast, our method

establishes a more generalized workflow for the robotic arm to manage material by determining two axes in the elevation view. A YOLO model was trained on a dataset featuring successful placements of timber elements in the elevation view. To collect this data, we filmed videos from the elevation view while a human iteratively built the structure. Humans simulated the robotic arm's branch-throwing motion by holding a branch vertically with two fingers and releasing it to let it fall freely into the gaps of the existing structure. A total of 146 videos, each lasting 5 to 10 minutes, were recorded. From these videos, 600 images of successful placements on structures in progress were extracted. The structural ends with potential gaps for the next placements were classified into three types: big_v, fork, and mini_v. The dataset was labeled with anchor boxes capturing the features of these three types of gaps (Figure 5).

The trained detection model identifies gaps within the branch structure from the elevation view, enabling it to predict the optimal placement for the next branch. After training, the recognition confidence scores range from approximately 0.67 to 0.84.

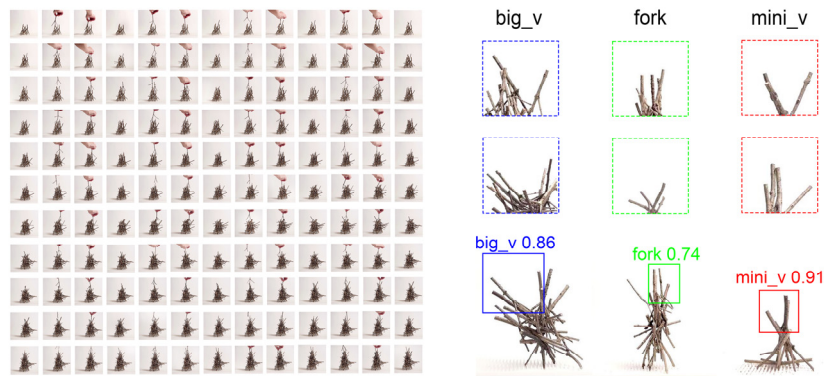


Figure 5 Structure gap dataset

4.4 INTERACTIVE ROBOTIC FABRICATION PROCESS

At the preparation stage, the participant presents a single timber branch to the robotic arm's camera. The robotic arm identifies the branch type and displays the category and detection box on a remotely connected computer. The participant then places the branch in the designated position within the preparation area. Subsequently, the participant uses 4 to 8 timber branches to build a foundation on the construction plate.

When the user testing begins, the robotic arm moves to position $x=4.00$ mm, $y=-70$ mm, and $z=88$ mm to view the entire structure. It then identifies the gap types using the trained model and displays the detection box and gap type on the screen of the remotely connected computer. If the recognition fails, the participant can add a timber branch and attempt the process again. Using computer vision (OpenCV team, 2021), the robotic arm calculates the center point of the detection box. After converting the camera coordinates to the robotic arm's coordinates, it moves to the center point's real-world position on the xz plane. If multiple detection boxes appear, it selects the one with the highest confidence score. The arm then raises its z -axis by 100 mm to avoid collision with the structure during subsequent movements. It also displays the required

branch type on the screen, which is pre-configured in the robotic arm's program based on prior human knowledge and successful construction experience.

In the next step, the participant guides the robotic arm to open the gripper to 95 degrees (Hiwonder, 2020) after it recognizes an 'OK' gesture through its camera. A branch matching the category previously indicated by the robotic arm is then placed by the participant in the gap of the gripper. The gripper is subsequently closed to 75 degrees (Hiwonder, 2020) when a 'fist' gesture is recognized. Through this process, the robotic arm successfully completes the task of holding a natural branch. Next, the system initiates a 2 cm forward movement when the 'two' gesture is recognized and moves backward by 2 cm when the 'one' gesture is recognized. In this way, the participant can control the position of the robotic arm along the y-axis. Once the participant judges that it has reached the optimal position, they guide the robotic arm to drop the branch vertically, allowing it to stack onto the current structure.

If the branch fails to stack on the existing structure, the participant can pick it up and allow the robotic arm to attempt the process again. If it fails a second time, the participant can change the branch type or dropping position and try once more. After a third failure, the stacking process is considered unsuccessful.

5. User Testing and Results

5.1 USER TESTING DESIGN

Ten participants were invited to test this interactive fabrication workflow. Among them, five had backgrounds in architectural design, one in industrial design, and four in engineering. Each participant was required to construct a timber structure five times. They collaborated with a robotic arm to drop timber branches onto the foundation, creating a dynamic design-build process. Before placing each branch, participants predicted its landing position, and after placement, the actual position was recorded. At the end of the testing, participants rated their confidence in the material and satisfaction with the design on a scale from -3 to 3, where -3 indicated extreme dissatisfaction or lack of confidence, 0 was neutral, and 3 indicated high satisfaction or strong confidence. They also answered questions about how the robotic arm assisted them during the fabrication process and shared their reflections on the overall workflow.

In this testing, participants are required to build a foundation as well as envision and review their designs to foster creativity. They collaborate with a robotic arm to drop branches, with the robotic arm providing input to promote innovation and help refine the structures at each step, creating a dynamic design-build process.

5.2 RESULT

The results of the user interviews and data analysis indicate that participants' confidence in the material generally increased as they completed more testings. (Figure 6) Some participants initially experienced a slight decrease in their confidence scores due to unsuccessful constructions, but their confidence improved afterward. This suggests that unsuccessful placements of timber elements heightened their awareness of geometric type choices. Over time, participants gradually understood the selection process and the functions of different timber branch types through trial and error.

Notably, most participants learned to adjust their branch type choices or select alternative drop locations if they encountered more than two unsuccessful placements. Additionally, participants generally felt more confident after the second round of creation, especially when design goals, such as vertical growth or maximizing cantilevers, remained consistent.

Throughout the testings, participants exhibited fluctuating levels of satisfaction with their designs. (Figure 6) While they noted that their structures became more stable as they gained a deeper understanding of materials through repeated trials, the outcomes were not always more satisfying. Satisfaction was higher when the robotic arm detected unexpected gaps, inspiring innovative and challenging designs. It also increased when the stacking process aligned with the participants' initial design intentions. All participants agreed that their decisions were critical to the final structure, influencing both its stability and the overall development of its form.

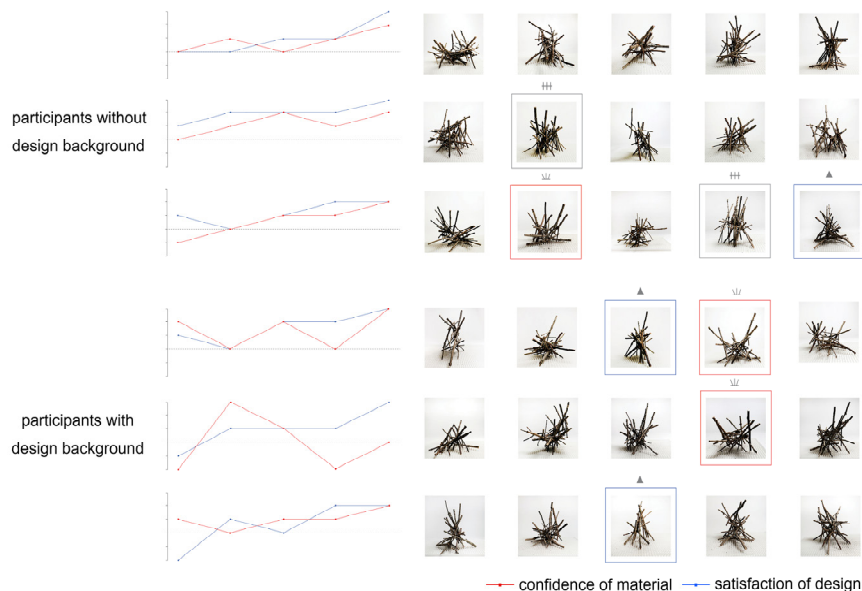


Figure 6 Part of the scores of material confidence and design satisfaction

The differences between the predicted position and the actual position of the timber recorded during the testing tended to decrease as participants conducted more trials. This indicates that participants gained a deeper understanding of the natural timber branches and began to account for the deviation between the robotic arm's drop and the ideal placement when designing. (Figure 7)

During the interviews, participants indicated that the detection capabilities of the robotic arm helped narrow down the range of possible placement positions and occasionally provided unexpected possibilities that inspired innovative design ideas. Additionally, they reflected that the limited freedom in gesture input during collaborative creation allowed them to focus more deeply on fabrication constraints and materiality. Moreover, participants expressed that the testings enhanced their

ability to utilize natural timber branches to achieve their design ideas and gradually understand how the forms of different branch types influenced the development of the structure. They also noted the limitations of each branch type during fabrication. Overall, reflecting on the entire process, participants gained a deeper understanding of the detection model, the robot's performance based on the existing structure, and the crucial role of material constraints in generating new forms.

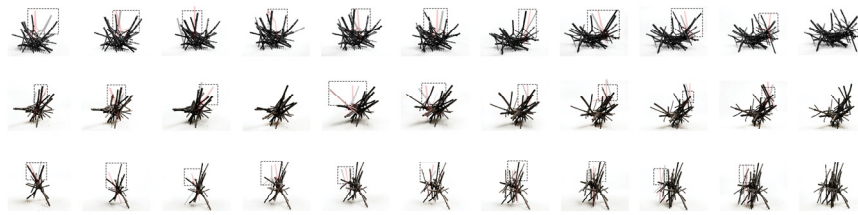


Figure 7 Comparison of predicted position and actual position in building process

6. Conclusion

This paper proposes an interactive workflow to explore the learning and fabrication of natural materials through human-machine cooperation. The workflow utilized a desktop robotic arm integrated with a camera to classify and detect categorized natural timber branches, stacking them into structural mockups. Two YOLOv5 models were trained to enable the robotic arm to classify branches and detect gaps within the structures. Within this workflow, humans provided y-axis input through gestures, while the robotic arm computed the x- and z-axis data using computer vision algorithms. The testings demonstrated the method's effectiveness in creating novel natural timber structures through interactive human-robot collaboration. The developed workflow can be adapted to accommodate other materials, enabling robotic pick-and-place tasks that balance human input and machine control, fostering creativity while maintaining structural stability.

To evaluate the application of the proposed design space and interactive fabrication method, 10 participants were invited to explore the design space, resulting in the creation of 50 structures. Participants were required to build timber structures based on the detection capabilities of the robotic arm and their understanding of the material. The results of the testings revealed that participants tended to develop a better understanding of the building potential and limitations of natural materials, as well as the detection capabilities of the robotic arm, with an increasing number of trials. The workflow could be further developed into a design tool to help users understand structural stability and the building logic inherent to natural materials.

However, it must be noted that the model training process is context-dependent and labor-intensive before the system becomes operational. The models need to be re-trained for each specific material type and form to achieve similar performance. Additionally, categorizing branches into only three types may oversimplify the complexity of the material, as the testings revealed that the angle between branches plays a critical role in the construction process. Enhanced classification capabilities for the robotic arm would improve performance. Furthermore, the robotic arm in the

testing relied on a single camera to perform multiple tasks, which participants found to be a less smooth experience. Lastly, the project's scale was limited by the robotic arm's capabilities, significantly impacting the structural performance of the resulting forms.

In this context, future work will explore other materials at larger scales to reflect more realistic constraints encountered in industrial construction with robots. Human input will be more precisely calibrated to adapt to complex material systems and fabrication conditions. Additionally, timber branches could be classified into more detailed categories, accounting for both the number and angles of branchings. Moreover, incorporating a top-down camera to monitor the structure's center of gravity may enable the robotic arm to autonomously complete the entire dropping process. By alternating turns between the robotic arm and human participants to drop branches, future studies could investigate whether this interaction yields novel and intriguing results.

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